Karollyn Larissa de Quadros

Environmental Engineer | Water Resources & Data Analytics Specialist
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Professional Summary

Environmental Engineer with a Master's degree in Environmental Engineering in its final stages and a background in engineering consultancy and applied research institutes. Professional experience focused on operational hydrology, water availability analysis, and drought management, delivering technical solutions for public water utilities and highway infrastructure sectors. Hands-on experience applying rainfall-runoff models (GR-type conceptual models, statistical regressions), hydroclimatic trend analysis, hydrometry, and quality control/data consistency for hydrometeorological sensor networks. Proficient in R and Python for data automation, climate data processing, and bias correction of atmospheric models. Proven track record in technical project management, preparing engineering reports for corporate clients, and liaising with institutional regulatory bodies.

Core Competencies

- **Technical Expertise:** Environmental Data Analytics, Hydrological Modeling, GIS & Spatial Analysis, Operational Monitoring.
- **Software & Tools:** R, Python, QGIS, Delft3D, LaTeX, MS Office Suite, Linux OS.
- **Key Capabilities:** Technical Report Writing, Environmental Risk Assessment, Client Communication, Quality Control.
- **Data Analytics & Formats:** Hydroclimatic Time Series Analysis, Spatial/Gridded Data Processing (NetCDF, GeoTIFF, Shapefiles), Data Cleansing & QA/QC, Automated Data Pipelines.
- **Hydrological Methods:** Extreme Value Analysis, Flood & Drought Frequency Analysis, Bias Correction, Grid Generation, Coordinate Reference Systems (CRS).

Professional Experience

Environmental Data Analyst / Applied Researcher – SIMEPAR Aug 2024 – Present

- **1st Place / Winner of the SIMEPAR Corporate Hackathon (2026):** Designed and prototyped an innovative solution for real-time monitoring and landslide risk assessment along highway slopes.
- Manage environmental and hydrological datasets to support operational monitoring systems for public and private clients.
- Developed a pilot early warning system for landslides based on critical rainfall accumulation threshold curves.
- Co-developed a regional drought status index used by state water supply utilities for resource management.
- Prepare monthly technical reports and maintain direct communication with corporate and government stakeholders.
- Generate routine operational products for water supply source monitoring.

Data Analyst Intern – SIMEPAR Sep 2023 – Aug 2024

- Applied R programming for automated processing, statistical analysis, and validation of large hydroclimatic datasets.
- Integrated remote sensing data (GRACE/GRACE-FO) into regional hydrological assessments.

Environmental Engineering Intern – Lactec Apr 2021 – Apr 2022

- Processed, validated, and assured quality control (QA/QC) for complex environmental datasets.
- Assisted in writing technical documentation and baseline reports for engineering projects.

Environmental Engineering Intern – RHA Engenharia e Consultoria Apr 2020 – Apr 2021

- Supported consulting teams in environmental impact studies and hydrological diagnosis.
- Performed data consistency checks and organized spatial and tabular data for engineering reports.

Education

M.Sc. in Environmental Engineering – PPGEA / UFPR Aug 2024 – Present

- Focus on basin-scale water balance, remote sensing integration, and applied hydrology.

B.Sc. in Environmental Engineering – UFPR Jan 2018 – Aug 2024

- Awarded Gold Medal for graduating 1st in class. Focus on numerical modeling and fluid mechanics.

Key Projects & Achievements

- **SIMEPAR Hackathon Winner (2026):** Awarded 1st place for designing an automated monitoring and early warning solution for highway slope stability and landslide risk.
- **UFPR Academic Excellence Award (2024):** Ranked 1 overall in Environmental Engineering graduating class (Gold Medal).
- **Hydrodynamic Lake Modeling (2024):** Modeled internal wave dynamics (Delft3D) to assess floating solar panel impacts.

Languages

Portuguese: Native | **English:** Advanced | **French:** Intermediate